

 rishill10@ASUS16X: ~/OS_LAB

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rishill10@ASUS16X:~$ mkdir OS_LAB
mkdir: cannot create directory 'OS_LAB': File exists
rishill10@ASUS16X:~$ cd OS_LAB
rishill10@ASUS16X:~/OS_LAB$ nano task1.c
rishill10@ASUS16X:~/OS_LAB$ gcc task1.c -o task1
rishill10@ASUS16X:~/OS_LAB$ ./task1
Enter a Linux command: ls
Parent Process
Parent PID : 693
Child PID  : 694
Child Process
Child PID : 694
Executing command: ls
task1  task1.c
Child process completed.
rishill10@ASUS16X:~/OS_LAB$ _
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rishill10@ASUS16X: ~/OS_LAB
task1 task1.c
Child process completed.
rishill10@ASUS16X:~/OS_LAB$ uname -a
Linux ASUS16X 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_
64 x86_64 x86_64 GNU/Linux
rishill10@ASUS16X:~/OS_LAB$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:                39 bits physical, 48 bits virtual
Byte Order:                   Little Endian
CPU(s):                       16
On-line CPU(s) list:         0-15
Vendor ID:                    GenuineIntel
Model name:                   12th Gen Intel(R) Core(TM) i5-12500H
CPU family:                   6
Model:                       154
Thread(s) per core:          2
Core(s) per socket:          8
Socket(s):                    1
Stepping:                     3
BogoMIPS:                     6220.78
Flags:                        fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clf
                              lush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_
                              good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmu
                              lqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline
                              _timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd
                              ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjus
                              t bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni x
                              saveopt xsavec xgetbv1 xsaves avx_vnni vnmi umip waitpkg gfni vaes vpclmulq
                              dq rdpid movdiri movdir64b fsrm md_clear serialize ibt flush_l1d arch_capab
                              ilities

Virtualization features:
Virtualization:               VT-x
Hypervisor vendor:            Microsoft
Virtualization type:          full
Caches (sum of all):
L1d:                           384 KiB (8 instances)
L1i:                           256 KiB (8 instances)
L2:                            10 MiB (8 instances)
L3:                            18 MiB (1 instance)
NUMA:
NUMA node(s):                  1
NUMA node0 CPU(s):             0-15
Vulnerabilities:
Gather data sampling:          Not affected
Ghostwrite:                    Not affected
Indirect target selection:     Not affected
Itlb multihit:                 Not affected
L1tf:                          Not affected
Mds:                           Not affected
Meltdown:                      Not affected
Mmio stale data:               Not affected
Old microcode:                 Not affected
Reg file data sampling:        Mitigation; Clear Register File
Retbleed:                      Mitigation; Enhanced IBRS

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Old microcode:          Not affected
Reg file data sampling: Mitigation; Clear Register File
Retbleed:               Mitigation; Enhanced IBRS
Spec rstack overflow:   Not affected
Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:             Mitigation; usercopy/swaps barriers and __user pointer sanitization
Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S

Srbds:                  Not affected
Tsa:                    Not affected
Tsx async abort:        Not affected
Vmscape:                Not affected

rishill10@ASUS16X:~/OS_LAB$ ps
  PID TTY          TIME CMD
  520 pts/0      00:00:00 bash
  701 pts/0      00:00:00 ps

rishill10@ASUS16X:~/OS_LAB$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1        0  0 07:13 ?           00:00:00 /sbin/init
root           2        1  0 07:13 hvc0       00:00:00 /init
root           7        2  0 07:13 hvc0       00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8
root          53        1  0 07:13 ?           00:00:00 /usr/lib/systemd/systemd-journald
root         102        1  0 07:13 ?           00:00:00 /usr/lib/systemd/systemd-udevd
root         113        1  0 07:13 ?           00:00:00 snapfuse /var/lib/snapd/snaps/tree_54.snap /snap/tree
root         114        1  0 07:13 ?           00:00:00 snapfuse /var/lib/snapd/snaps/core22_2411.snap /snap
root         119        1  0 07:13 ?           00:00:03 snapfuse /var/lib/snapd/snaps/snapd_27591.snap /snap
systemd+     141        1  0 07:13 ?           00:00:00 /usr/lib/systemd/systemd-resolved
systemd+     150        1  0 07:13 ?           00:00:00 /usr/lib/systemd/systemd-timesyncd
root         210        1  0 07:13 ?           00:00:00 /usr/sbin/cron -f -P
message+     211        1  0 07:13 ?           00:00:00 @dbus-daemon --system --address=systemd: --nofork --
root         217        1  0 07:13 ?           00:00:00 /snap/snapd/current/usr/lib/snapd/snapd
root         218        1  0 07:13 ?           00:00:00 /usr/lib/systemd/systemd-logind
root         239        1  0 07:13 tty1       00:00:00 /sbin/agetty -o -p -- \u --noclear - linux
syslog       241        1  0 07:13 ?           00:00:00 /usr/sbin/rsyslogd -n -iNONE
root         251        1  0 07:13 ?           00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unat
root         385        2  0 07:13 pts/1      00:00:00 /bin/login -f
rishill+     443        1  0 07:13 ?           00:00:00 /usr/lib/systemd/systemd --user
rishill+     444       443  0 07:13 ?           00:00:00 (sd-pam)
rishill+     468       385  0 07:13 pts/1      00:00:00 -bash
root         515        2  0 07:14 ?           00:00:00 /init
root         516       515  0 07:14 ?           00:00:00 /init
rishill+     520       516  0 07:14 pts/0      00:00:00 -bash
root         702       102  0 07:35 ?           00:00:00 (udev-worker)
root         703       102  0 07:35 ?           00:00:00 (udev-worker)
rishill+     704       520  0 07:35 pts/0      00:00:00 ps -ef

rishill10@ASUS16X:~/OS_LAB$ top
top - 07:36:16 up 23 min,  1 user,  load average: 0.01, 0.02, 0.00
Tasks: 27 total,  1 running, 26 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7773.7 total, 7091.3 free,  594.7 used,  239.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,   0.0 used. 7179.0 avail Mem

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root      702      102    0 07:35 ?        00:00:00 (udev-worker)
root      703      102    0 07:35 ?        00:00:00 (udev-worker)
rishill+  704      520    0 07:35 pts/0    00:00:00 ps -ef
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rishill10@ASUS16X:~/OS_LAB$ top
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top - 07:36:28 up 23 min,  1 user,  load average: 0.01, 0.02, 0.00
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Tasks:  25 total,   1 running, 24 sleeping,   0 stopped,   0 zombie
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```
%Cpu(s):  0.0 us,   0.0 sy,   0.0 ni,100.0 id,   0.0 wa,   0.0 hi,   0.0 si,   0.0 st
```

```
MiB Mem :  7773.7 total,  7095.1 free,   590.9 used,   239.2 buff/cache
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```
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7182.8 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21744	13020	9636	S	0.0	0.2	0:00.66	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
7	root	20	0	3196	2132	2008	S	0.0	0.0	0:00.00	init
53	root	19	-1	50372	16152	15080	S	0.0	0.2	0:00.19	systemd-journal
102	root	20	0	25508	6852	5148	S	0.0	0.1	0:00.61	systemd-udev
113	root	20	0	153076	1876	1552	S	0.0	0.0	0:00.00	snapfuse
114	root	20	0	153076	1812	1488	S	0.0	0.0	0:00.00	snapfuse
119	root	20	0	452000	12220	1512	S	0.0	0.2	0:03.71	snapfuse
141	systemd+	20	0	21344	13160	11036	S	0.0	0.2	0:00.11	systemd-resolve
150	systemd+	20	0	91036	7944	6972	S	0.0	0.1	0:00.10	systemd-timesyn
210	root	20	0	4244	2688	2432	S	0.0	0.0	0:00.00	cron
211	message+	20	0	9604	5364	4676	S	0.0	0.1	0:00.05	dbus-daemon
217	root	20	0	2294224	37836	24576	S	0.0	0.5	0:00.85	snappd
218	root	20	0	17972	8868	7828	S	0.0	0.1	0:00.07	systemd-logind
239	root	20	0	3124	1976	1836	S	0.0	0.0	0:00.00	agetty
241	syslog	20	0	222516	5968	4588	S	0.0	0.1	0:00.06	rsyslogd
251	root	20	0	107036	23264	13808	S	0.0	0.3	0:00.18	unattended-upgr
385	root	20	0	6704	4652	3864	S	0.0	0.1	0:00.01	login
443	rishill+	20	0	20384	11604	9452	S	0.0	0.1	0:00.13	systemd
444	rishill+	20	0	21172	3576	1848	S	0.0	0.0	0:00.00	(sd-pam)
468	rishill+	20	0	6080	5372	3708	S	0.0	0.1	0:00.02	bash
515	root	20	0	3184	1120	980	S	0.0	0.0	0:00.00	SessionLeader
516	root	20	0	3200	1260	1108	S	0.0	0.0	0:00.03	Relay(520)
520	rishill+	20	0	6080	5364	3648	S	0.0	0.1	0:00.05	bash
706	rishill+	20	0	9328	5712	3500	R	0.0	0.1	0:00.02	top